

base disturbance w_{ext} (ACS/thruster-noise proxy)



Free-floating environment (MuJoCo)
chaser base + planar 3-DOF arm
+ tumbling non-cooperative target
compliant contact, zero- g

obs $s \in \mathbb{R}^{32}$ (kinematics only)



SAC policy π
[256, 256] MLP



torque τ (± 5 N·m)



reward r (Eq. 1)